

acquiring and building a new customer takes a lot of time and energy. We also offer excellent pre-sales and post-sales support. In defence, even before we get the orders, we start the design work as the paperwork may take a long time."

Long-term partnerships foster growth. Sudhir Naik, vice president-corporate affairs, eInfochips, said, "Post-recession, the thought process of customers has changed. They focus no more on cost arbitration but long-term partnership."

Domain expertise. Customers prefer design houses that have strong expertise in specific verticals. Vasudevan said, "We have deep domain expertise to do complex designs both at silicon level and product level. We have undertaken and delivered successfully many high-complexity designs in telecom, computing and other industries. With our deep expertise, we are able to manage high-complexity product designs better. Also, customers want us to take up complete ownership of product design and engineering."

Cherukuri informed, "Infotech's cutting edge is industry-specific domain expertise, people and processes, technologies, tools and training. We draw on this strong foundation to create measurable business impact for customers around the world, resulting in long-term relationships with several

Technology trends

Low power and geometry. With the need for smaller chips, there is a push towards designing in lower geometries. Hence understanding the challenges in designing small chips with high density is crucial. Also, there is a trend towards use of multi-core and ultra-low power consumption.

Green technology. Green and clean technology is becoming important, and it is manifested differently in different industries. From designing new engines with low noise and fuel consumption to designing lead-free products, there is a lot of scope.

Integration. Raveendran Nair, CEO, Educe Micro Research, revealed, "Multiple functions (power, analogue, intelligence and communication)—compared to any one previously—are being integrated."

Software-based customisation. Shankar pointed out another trend: "We are seeing a greater consolidation of hardware platforms and customisation through software. Also, there are more features in the same chip which can be enabled or disabled through software. The same chip can be used in different cellphones, whether the handset is priced at Rs 1500 or Rs 20,000, as software customisation helps to make the chip work differently with different functions."

Vasudevan added, "Feature variation, especially using software variants, is increasing. In the semiconductor industry, the trend is towards system LSI and complete system-on-chip. From full-system functionality, understanding the end-market and product needs is also important."

Device convergence. Ramakrishnan said, "Device convergence is a major trend that is influencing semiconductor chip design. Power management and high-speed communications are important for it. We are seeing an increased demand for high-efficiency, low-power chips for consumer electronics and mobile applications, niche automotive applications (for hybrid and electric vehicles), alternative energy applications, etc."

of the most recognised names in their respective industries."

Marketing. For acquiring new clients, 90 per cent of the surveyed IDHs participated in events. Seventy-five per cent also believed in one-on-one meetings, followed by word-of-mouth publicity (70 per cent). Other major marketing tools adopted were speaking at events (55 per cent), e-mailers

(40 per cent), Web advertising and personal contacts (25 per cent each), international silicon conferences (20 per cent) and print ads, Google ads, technical articles and own websites (10 per cent each). Some also looked upon SMS blasts, industry forums, customer events, and tie-ups with industry consortiums and research institutes.

Visweswaran B., CEO, Wysine Technologies, said, "We rely on word-of-mouth and one-on-one meetings, and do use e-mailers. We are also working on creating a channel by partnering with other companies in the energy ecosystem—such as UPS or diesel generator manufacturers and HVAC system integrators. Our participation in events organised by CII-IGBC and TERI also helps in creating awareness about our products. Lately, we have started using social networking sites too."

Meeting manpower needs

The surveyed IDHs employed design engineers with experience ranging between one and 15 years, with a majority of them preferring those with four to five years of experience.

Marketing Tools

